

D3

$$a_n = \frac{4^n - 6^n}{n3^n - 5^n} \sim \frac{-6^n}{-5^n} = \frac{6^n}{5^n} = +\infty$$

D6

$$y = x^2 e^{x-2}$$

$$y' = (2x)(e^{x-2}) + (x^2)(e^{x-2}) = 2x e^{x-2} + x^2 e^{x-2} = e^{x-2}(2x + x^2)$$

$$e^{x-2}(2x + x^2) \Rightarrow 2x + x^2 < 0 \Rightarrow x(2+x) < 0 \begin{cases} x < 0 \\ 2+x < 0 \end{cases} \begin{cases} x < 0 \\ x < -2 \end{cases}$$

$$x \in (-2, 0)$$

D8

$$\int_0^2 \frac{x}{x^2+1} = \left[\frac{1}{2} \ln(x^2+1) \right]_0^2 = \left(\frac{1}{2} \ln(4+1) \right) - \left(\frac{1}{2} \ln(1) \right) = \left(\frac{1}{2} \ln(5) \right) - 0 = \frac{\ln(5)}{2}$$

$$\int \frac{x}{x^2+1} = \int \frac{\cancel{x}}{x^2+1} \cdot \frac{1}{\cancel{2x}} = \frac{1}{2} \int \frac{1}{y} = \frac{1}{2} \ln(y) + C = \frac{1}{2} \ln(x^2+1) + C$$

$$y = x^2 + 1$$

$$dy = 2x dx$$

$$\frac{dy}{2x} = dx$$

D2

$$-\sqrt{x^4 - 2x^2 + 1} = -\sqrt{(x^2-1)^2} = -|x^2-1|$$

D7

$$y = x^2 \ln(x)$$

$$y' = (2x)(\ln(x)) + \left(x^2 \cdot \frac{1}{x} \right) = 2x \ln(x) + x$$

$$y'' = (2)(\ln(x)) + \left(2x \cdot \frac{1}{x} \right) = 2 \ln(x) + 2 + 1 = 2 \ln(x) + 3$$

$$2 \ln(x) + 3 < 0 \Rightarrow \ln(x) < -\frac{3}{2} \Rightarrow x < e^{-3/2} \Rightarrow x < \frac{1}{\sqrt{e^3}}$$

$$x \in \left(0, \frac{1}{\sqrt{e^3}} \right)$$

D5

$$y = \begin{cases} \frac{e^x - 1}{2^x} & x < 0 \\ 2^{k+x} & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} \frac{e^0 - 1}{2 \cdot 0} = \frac{0}{0} \text{ l'Hôpital } \frac{e^x}{2} = \frac{1}{2}$$

$$2^{k+0} = \frac{1}{2} \Rightarrow 2^k = \frac{1}{2} \Rightarrow 2^k = 2^{-1} \Rightarrow k = -1$$

D4

$$\sum_{n=1}^{\infty} \frac{(n+1)^n}{(3n)^{2n}} \quad \frac{\sqrt[n]{(n+1)^n}}{\sqrt[n]{(3n)^{2n}}} = \frac{n+1}{9n^2} \sim \frac{n}{9n^2} = 0 \quad \text{la serie converge}$$

E1

$$y = \frac{\sqrt{4x^2 + 1}}{x^2 + 1}$$

DOMINIO

$$x \in \mathbb{R}$$

LIMITI

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^2 + 1}}{x^2 + 1} = 0 \quad \text{perche' asintota orizzontale}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 1}}{x^2 + 1} = 0 \quad \text{perche' asintota orizzontale}$$

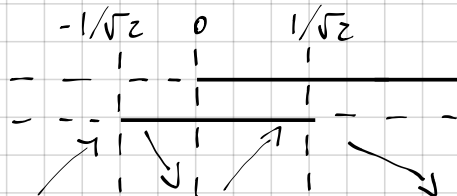
MONOTONIA

$$y = \frac{\sqrt{4x^2 + 1}}{x^2 + 1}$$

$$y' = \left(\frac{1}{2} \cdot \frac{1}{\sqrt{4x^2 + 1}} \cdot 8x \right) (x^2 + 1) - \left(\sqrt{4x^2 + 1} \right) (2x) = \left(\frac{4x}{\sqrt{4x^2 + 1}} \right) (x^2 + 1) - (2x \sqrt{4x^2 + 1})$$

$$\left(\frac{4x^3 + 4x}{\sqrt{4x^2 + 1}} \right) - \frac{2x \sqrt{4x^2 + 1}}{(x^2 + 1)^2} = \frac{4x^3 + 4x - 2x(4x^2 + 1)}{(x^2 + 1)^2 \sqrt{4x^2 + 1}} = \frac{4x^3 + 4x - 8x^3 - 2x}{(x^2 + 1)^2 \sqrt{4x^2 + 1}} = \frac{-4x^3 + 2x}{(x^2 + 1)^2 \sqrt{4x^2 + 1}}$$

$$-4x^3 + 2x > 0 \Rightarrow 2x(-2x^2 + 1) \begin{cases} 2x > 0 \\ -2x^2 + 1 > 0 \end{cases} \begin{cases} x > 0 \\ 2x^2 - 1 < 0 \end{cases} \begin{cases} x > 0 \\ x^2 - \frac{1}{2} < 0 \end{cases} \begin{cases} x > 0 \\ x^2 < \frac{1}{2} \end{cases} \begin{cases} x > 0 \\ -1/\sqrt{2} < x < 1/\sqrt{2} \end{cases}$$



↗ crescente su $(-\infty, -1/\sqrt{2}) \cup (0, 1/\sqrt{2})$

↘ decrescente su $(-1/\sqrt{2}, 0) \cup (1/\sqrt{2}, +\infty)$

MIN & MAX

$$y' = \frac{4x^3 + 2x}{(x^2 + 1)^2 \sqrt{4x^2 + 1}}$$

$$\frac{4x^3 + 2x}{(x^2 + 1)^2 \sqrt{4x^2 + 1}} = 0 \quad 4x^3 + 2x = 0 \quad 2x(2x^2 + 1) = 0 \quad x = \pm 1/\sqrt{2}$$

$$x = 0$$

$$x = \pm 1/\sqrt{2}$$

$$f(x) = \frac{\sqrt{4x^2 + 1}}{x^2 + 1}$$

$$f(0) = \frac{\sqrt{4 \cdot 0 + 1}}{1} = 1$$

$$f(-1/\sqrt{2}) = \frac{\sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2 \cdot 4 + 1}}{\left(-\frac{1}{\sqrt{2}}\right)^2 + 1} = \frac{\sqrt{\frac{4}{2} + 1}}{\frac{1}{2} + 1} = \frac{\sqrt{3}}{\frac{3}{2}} = \sqrt{3} \cdot \frac{2}{3} = \frac{2\sqrt{3}}{3}$$

$$f(1/\sqrt{2}) = \frac{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 \cdot 4 + 1}}{\left(\frac{1}{\sqrt{2}}\right)^2 + 1} = \frac{\sqrt{\frac{4}{2} + 1}}{\frac{1}{2} + 1} = \frac{\sqrt{3}}{\frac{3}{2}} = \sqrt{3} \cdot \frac{2}{3} = \frac{2\sqrt{3}}{3}$$

$f(\pm 1/\sqrt{2})$ massimi globali

g2

Il teorema del confronto per serie afferma che, dati 2 successioni a_n e b_n a termini positivi se $a_n \geq b_n \geq 0$

o a_n converge o b_n converge, o b_n diverge o a_n diverge

2)

$$\sum \frac{\sin(n^2) + 2}{n^2 + n}$$
$$0 \leq \frac{\sin(n^2) + 2}{n^2 + n} < \frac{3}{n^2}$$

$\frac{3}{n^2}$ come essendo una serie armonica con esponente $2 > 1$

quindi $\sum a_n$ converge

3)

$$\sum \frac{\sin(n^2) + 2}{n+1}$$

$$\frac{\sin(n^2) + 2}{n+1} \geq \frac{1}{n+1} \geq 0$$

$\frac{1}{n+1}$ è una serie armonica che diverge essendo che l'esponente $0 \leq 1$, quindi anche $\sum a_n$ diverge